

Algebra First-Year Practice Exam B, Part 2

Answer four problems. Write your answers clearly in complete English sentences. You may quote results (within reason) as long as you state them clearly.

1. This problem has three parts.
 - (a) Prove that a prime $p \in \mathbb{N}$ such that $p \equiv 3 \pmod{4}$ is prime when considered as an element in the ring of Gaussian integers $\mathbb{Z}[i]$.
 - (b) Prove that a prime $p \in \mathbb{N}$ such that $p \equiv 1 \pmod{4}$ is a product of two non-associate primes in $\mathbb{Z}[i]$.
 - (c) Give a factorization of 30 as the product of a unit and powers of non-associate primes in $\mathbb{Z}[i]$.
2. Let R be the ring of $n \times n$ matrices over a field F , where $n \geq 2$.
 - (a) Show that R has no ideals other than 0 and R .
 - (b) Exhibit a nonzero proper left ideal of R .
3. Find one representative of each conjugacy class of elements of order 4 in the group $GL(6, \mathbb{Q})$.
4. Let F be a field.
 - (a) What does it mean to say that a field extension of F is algebraic?
 - (b) Let $F \subset K \subset L$ be three fields. Prove that if K is algebraic over F and L is algebraic over K , then L is algebraic over F .
5. A commutative ring R with 1 is called a local ring if and only if it has a unique maximal ideal. Suppose that R is a ring with a proper ideal M . Show that R is local, with maximal ideal M , if and only if every element of $R \setminus M$ (set difference) is a unit.